

Metadata Standardization and Validation Efforts by IGS for VMRC Data

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A. Introduction to VMRC metadata

Metadata Standards

Adoption of MlxS standard, defined by Genomic Standards Consortium (GSC), is the best approach to report metadata which extends the minimum information already captured by INSDC or any other data repository. GSC has also introduced 'environmental packages' that standardize sets of measurements and observations describing the environment from which a biological sample originates. Using MlxS checklist along with the 'vaginal environment package', will ensure and elevate the quality, accessibility and utility of contextual information related to VMRC whole metagenome and 16S amplicon datasets.

MIMAG standard

MIMAG, or Minimum Information about a Metagenome-Assembled Genome, is a standard developed by Genomic Standards Consortium (GSC) for reporting metadata about bacterial and archaeal genome sequence assemblies resulting from metagenomic datasets. MIMAG is an extension of the Minimum Information about Any (x) Sequence (MlxS) standard.

More information on the MIMAG standard is available here:

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6436528/>

MIMARKS standard

MIMARKS, or Minimum Information about a marker gene sequence, is a standard developed by the Genomic Standards Consortium (GSC) for reporting metadata about marker gene sequences. MIMARKS is an extension of the Minimum Information about Any (x) Sequence (MlxS).

More information on MIMARKS standard is available here:

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3367316/>

VMRC Metadata Templates

MIMAG and MIMARKS metadata

For the VMRC project, we use two different metadata templates for capturing file-, sample-, and subject-level information for the VMRC datasets. The MIMAG metadata template is designed specifically for metagenomics, metatranscriptomics, isolate genome, metabolome, metaproteome, immunology and RNA-seq data types. The MIMARKS metadata template is tailored for 16S and ITS amplicon data types. These templates are modifications of the standards described above in which we have added additional metadata fields from other sources like NCBI's BioSample, the IGS VMRC portal, and the MlXS vaginal environment checklists, Espinosa (ZAPPS/PreSSMat) dataset, NCBI's GEO data repository and NEMO data archive.

Study-level metadata

We provide a metadata template which captures study-level information. The source of attributes used for this metadata is the [STORMS](#) (Strengthening The Organization and Reporting of Microbiome Studies) checklist. The STORMS checklist provides guidance for concise and complete reporting of microbiome studies that will facilitate manuscript preparation, peer review, and reader comprehension of publications and comparative analysis of published results.

Summary-files metadata

We also provide a metadata template which captures file-level information of all the summary files from other types of analyses, like immune protein profiling, metabolite profiling etc.

Controlled Vocabulary (CV) file

We also provide a Controlled Vocabulary file, which can be used as a reference to identify the ontology (EDAM,OBI,UBERON) ids for data/file types or other terms . This helps to facilitate use of ontology terms, eliminate free-text entry whenever possible, and to ensure standard vocabulary for a set of terms. Here are the links to browse these ontologies.

EDAM - An ontology of well established, familiar concepts that are prevalent within bioinformatics, including types of data and data identifiers, data formats, operations and topics. <https://www.ebi.ac.uk/ols4/ontologies/edam>

OBI - An ontology for representing biomedical investigations, including study designs, the collection and preparation of the targets of investigation, assays, instrumentation and reagents used, as well as the data generated and the types of analysis performed on the data to reach conclusions, and their documentation. <https://www.ebi.ac.uk/ols4/ontologies/obi>

UBERON - An integrated cross-species anatomy ontology representing a variety of entities classified according to traditional anatomical criteria such as structure, function and developmental lineage. The ontology includes comprehensive relationships to taxon-specific anatomical ontologies, allowing integration of functional, phenotype and expression data. <https://www.ebi.ac.uk/ols4/ontologies/uberon>

CL (Cell Ontology) - An ontology designed to classify and describe cell types across different organisms. It serves as a resource for model organism and bioinformatics databases. <https://www.ebi.ac.uk/ols4/ontologies/cl>

NCBI Taxonomy Browser - The taxonomy database maintained by NCBI/GenBank, which allows browsing of the taxonomy tree, which contains a classification of organisms. <https://www.ncbi.nlm.nih.gov/Taxonomy/Browser/wwwtax.cgi>

Links to metadata templates and attribute description files

Below are the Google Drive links to the templates and attribute informational files for MIMAG, MIMARKS, study-level metadata and summary-file metadata.

[MIMAG metadata template file with attribute descriptions](#)

[MIMARKS \(16S/ITS\) metadata template file with attribute descriptions](#)

[Study metadata template file with attribute descriptions](#)

[Summary files metadata template file with attribute descriptions](#)

[Controlled vocabulary file](#)

VMRC metadata attributes

Both MIMAG and MIMARKS metadata files capture file, sample and subject-level information about our datasets. Both metadata templates have many common attributes ([Table 1](#)), as well as some attributes which are unique to MIMAG-specific ([Table 2](#)) or MIMARKS-specific ([Table 3](#)) data types. Refer to the Tables for a short description and example for the metadata attributes.

The last column in the table denotes if a value in that field is required or optional according to the following key:

Optional - X - this field can be left blank

Mandatory - M - a value must be entered for this field

Conditional Mandatory - C - this field is mandatory when certain conditions are met

Table 1. Attributes common to MIMAG and MIMARKS metadata file
{Fields marked with asterisk (*) are also found in summary files metadata}

Attribute	Group	Description	Example(s)	Mandatory(M) /Optional(X) /Conditional Mandatory (C)
project_name*	Database_Identifiers	Name of the project within which the sequencing was organized	VMRC	M
study_name*	Database_Identifiers	Name of the study, in the format of 'project_name'_'study'	VMRC_PreSS Mat	M
bioproject_accession	Database_Identifiers	The BioProject accession number assigned to your project	PRJNAXXXXXX X	X
file_name*	File_Information	Name of the file in gzipped format for reads, assemblies etc. For metagenomic sequence reads and assemblies, use prefix "MG_", for Metatranscriptomic sequence reads and assemblies, use prefix "MT_". For 16S data files, include the variable region studied (v4.R1.fastq.gz) <i>Note : This field should be unique throughout.</i>	MG_20_00880_05.R1.fq.gz, 20-00145-05.v4.R2.fastq.gz	M
file_desc*		{Only applies to summary files metadata.} Provide a short description of what the summary file data is about.	Metabolomic profiling, Immune Protein profiling via Luminex, Metagenomic reads mapped to VIRGO annotation table, 16S abundance profile with ASVs and taxonomy information	M
uncompressed_filesize*	File_Information	Size of file in bytes for uncompressed file		X
compressed	File_Information	Size of file in bytes for		X

_filesize	ation	compressed/gzipped file		
uncompress ed_file_md5 *	File_Inform ation	md5sum for file in uncompressed format		M
compressed _file_md5	File_Inform ation	md5sum for file in compressed/gzipped format		M
access	File_Inform ation	Consented access level of data. Refer to the Controlled Vocabulary file. <i>Note: This field is mandatory. Enter Not Available, if information is not available.</i>	open	M
data_use_co ndition	File_Inform ation	Term from the Data Use Ontology (DUO) to describe how the data is allowed to be used. Please fill this field if the access level of your data is restricted or restricted_embargo. Refer to the Controlled Vocabulary file. https://www.ebi.ac.uk/ols4/ontologies/duo	Open data: DUO:0000004	C
file_type*	File_Inform ation	File format type and EDAM ontology id. Refer to the Controlled Vocabulary file.	fastq,EDAM:19 30	M
data_catego ry*	File_Inform ation	Data category of file and EDAM ontology id. Refer to the Controlled Vocabulary file.	Taxonomy,EDA M:3028	M
modality*	Sample_Inf ormation	Modality/assay information. Refer to the Controlled Vocabulary file.	16Samplicon, Metagenome	M
samp_taxon omy	Sample_Inf ormation	Sample taxonomy and NCBI taxonomy id information. Refer to the Controlled Vocabulary file. For Isolate genomes, provide sample taxonomy like Prevotella_bivia; for human microbial metagenomics, metatranscriptomics and 16Samplicon data, fill in as 'human vaginal metagenome,NCBI:txid1632839'.	Prevotella_bivi a,NCBI:txid281 25	C - Mandatory for Isolate genomes
strain	Sample_Inf ormation	Isolate/strain information for Isolate genomes. <i>This field is mandatory for Isolate genomes</i>		C - Mandatory for Isolate genomes
samp_name	Sample_Inf	A local identifier or name for the material	VMRC_PreSS	M

	ormation	sample used for extracting nucleic acids, and subsequent sequencing. It can refer either to the original material collected or to any derived sub-samples. Format recommended : 'study_name'_'samp_title'	Mat_20-00135-05	
samp_title	Sample_Inf ormation	A unique identifier for the sample, assigned by your lab	20-00135-05	M
samp_type	Sample_Inf ormation	Specify the type of your sample. Provide UBERON ontology term and ID, if available. Pick a term from the controlled vocabulary. Please contact us if you need help finding UBERON ontology ID for your sample type. https://www.ebi.ac.uk/ols4/ontologies/uber/classes/http%253A%252F%252Fpurl.obolibrary.org%252Fobo%252FUBERON_0000463	vaginal fluid,UBERON:0036243	X
cell_line	Sample_Inf ormation	Name of specific cell line used in your study. Refer to the Controlled Vocabulary file.	primary vaginal epithelial cells (PVECs)	C
cell_type	Sample_Inf ormation	Specify the CL ontology term and ID for the cell type of your sample. Please fill this field if you have provided cell line information. Refer to the Controlled Vocabulary file. Please contact us if you need help finding CL ontology term and ID for your specific cell type. https://www.ebi.ac.uk/ols4/ontologies/cl/classes/http%253A%252F%252Fpurl.obolibrary.org%252Fobo%252FCL_0000000?lang=en	vagina squamous cell,CL:1001578	C - Mandatory if cell_line information is provided
experimental_factor	Sample_Inf ormation	Experimental factors are essentially the variable aspects of an experiment design which can be used to describe an experiment, or set of experiments, in an increasingly detailed manner. This field accepts ontology terms from Experimental Factor Ontology (EFO) and/or Ontology for Biomedical Investigations (OBI). For a browser of EFO terms, please see https://www.ebi.ac.uk/ols4/ontologies/efo ; for a browser of OBI terms please see https://www.ebi.ac.uk/ols4/ontologies/obi	time series design [EFO:EFO_0001779]	X
lat_lon	Sample_Inf ormation	The geographical origin of the sample as defined by latitude and longitude. The	50.586825 6.408977	M

		values should be reported in decimal degrees and in the WGS84 system. <i>Note: This field is mandatory. Enter Not Available, if information is not available.</i>		
geo_loc_name	Sample_Information	The geographical origin of the sample as defined by the country or sea name followed by specific region name. Country or sea names should be chosen from the INSDC country list (http://insdc.org/country.html), or the GAZ ontology (http://purl.bioontology.org/ontology/GAZ) <i>Note: This field is mandatory. Enter a null value term from the Controlled Vocabulary file, if information is not available.</i>	USA: Maryland	M
site	Sample_Information	An identifier representing the site of study {AMANHIB = AMANHI Bangladesh, AMANHIP = AMANHI Pakistan, AMANHIT = AMANHI Tanzania, GAPPSB = GAPPS Bangladesh, GAPPSZ = GAPPS Zambia}	GAPPSB	X
collection_year	Sample_Information	Year of sample collection in the format of YYYY	2025	C
neg_cont_type	Sample_Information	The substance or equipment used as a negative control in an investigation		X
pos_cont_type	Sample_Information	The substance, mixture, product, or apparatus used to verify that a process which is part of an investigation delivers a true positive.		X
samp_collection_device	Sample_Information	The instrument or tool used to obtain a biological or environmental sample from its source. Refer to the Controlled vocabulary file.	swab	X
samp_collection_method	Sample_Information	Specify the technique or procedure used to obtain a sample from its source. Provide OBI ontology term and ID, if available. Refer to the Controlled Vocabulary file. Please contact us if you need help finding OBI ontology ID for the sample collection method used.	collecting specimen with swab,OBI:0002600	X

		https://www.ebi.ac.uk/ols4/ontologies/obi/classes/http%253A%252F%252Fpurl.obolibrary.org%252Fobo%252FOBI_0000659?lang=en		
samp_storage_medium	Sample_Information	The substance or condition in which a collected sample is stored before analysis. Refer to the Controlled Vocabulary file.	dry swab, nucleic acid stabilizing solution, Amies transport medium	X
samp_storage_method	Sample_Information	Describe the method used to preserve and store biological samples after collection to maintain their integrity for downstream analysis.	swab stored in DNA/RNA stabilizers and kept frozen at -80 degrees celsius ; in sterile transport medium at 4 degrees celsius; air-dried cytobrush stored in a sterile container	X
samp_store_temp	Sample_Information	Temperature at which sample was stored, e.g. -80 degree Celsius	-80 degree Celsius	X
samp_store_loc	Sample_Information	Location at which sample was stored, usually name of a specific freezer/room	Freezer no:5	X
samp_store_dur	Sample_Information	Duration for which the sample was stored	P1Y6M	X
salinity	Sample_Information	The total concentration of all dissolved salts in a liquid or solid sample. While salinity can be measured by a complete chemical analysis, this method is difficult and time consuming. More often, it is instead derived from the conductivity measurement. This is known as practical salinity. These derivations compare the specific conductance of the sample to a salinity standard such as seawater.	25 practical salinity unit	X

oxy_stat_samp	Sample_Information	Oxygenation status of sample	aerobic	X
ref_biomaterial	Sample_Information	Primary publication if isolated before genome publication; otherwise, primary genome report.	doi:10.1016/j.syapm.2018.01.009	X
source_mat_id	Sample_Information	A unique identifier assigned to a material sample (as defined by http://rs.tdwg.org/dwc/terms/materialSampleID , and as opposed to a particular digital record of a material sample) used for extracting nucleic acids, and subsequent sequencing. The identifier can refer either to the original material collected or to any derived sub-samples. The INSDC qualifiers /specimen_voucher, /bio_material, or /culture_collection may or may not share the same value as the source_mat_id field. For instance, the /specimen_voucher qualifier and source_mat_id may both contain 'UAM:Herps:14', referring to both the specimen voucher and sampled tissue with the same identifier. However, the /culture_collection qualifier may refer to a value from an initial culture (e.g. ATCC:11775) while source_mat_id would refer to an identifier from some derived culture from which the nucleic acids were extracted (e.g. xatc123 or ark:/2154/R2).	MPI012345	C - for culture collection
rel_to_oxygen	Sample_Information	Is this organism an aerobe, anaerobe? Please note that aerobic and anaerobic are valid descriptors for microbial environments	aerobe	X
samp_mat_process	Sample_Information	A brief description of any processing applied to the sample during or after retrieving the sample from the environment, or a link to the relevant protocol(s) performed.	storing samples in ethanol	X
size_frac	Sample_Information	Filtering pore size used in sample preparation	0-0.22 micrometer	X
samp_size	Sample_Information	The total amount or size (volume (ml), mass (g) or area (m2)) of sample collected.	5 liter	X
samp_vol_w_dna_ext	Sample_Information	Volume (ml) or mass (g) of total collected sample processed for DNA extraction. Note: total sample collected should be entered under the term Sample Size	1500 milliliter	X

		(MIXS:0000001).		
nucl_acid_ext	Sample_Information	A link to a literature reference, electronic resource or a standard operating procedure (SOP), that describes the material separation to recover the nucleic acid fraction from a sample		X
nucl_acid_amp	Sample_Information	A link to a literature reference, electronic resource or a standard operating procedure (SOP), that describes the enzymatic amplification (PCR, TMA, NASBA) of specific nucleic acids	https://phylogenomics.me/protocols/16s-pcr-protocol/	X
lib_size	Sample_Information	Total number of clones in the library prepared for the project	50	X
lib_reads_seqd	Sample_Information	Total number of clones sequenced from the library	20	
lib_layout	Sample_Information	Specify whether to expect single, paired, or other configuration of reads	paired	X
lib_vector	Sample_Information	Cloning vector type(s) used in construction of libraries	Bacteriophage P1	X
lib_screen	Sample_Information	Specific enrichment or screening methods applied before and/or after creating libraries	enriched, screened, normalized	X
mid	Sample_Information	Molecular barcodes, called Multiplex Identifiers (MIDs), that are used to specifically tag unique samples in a sequencing run. Sequence should be reported in uppercase letters	GTGAATAT	C-Recommended for 16S data
adapters	Sample_Information	Adapters provide priming sequences for both amplification and sequencing of the sample-library fragments. Both adapters should be reported; in uppercase letters	AATGATACGG CGACCACCG AGATCTACAC GCT; CAAGCAGAA GACGGCATAC GAGAT	X
omics_technique	Sample_Information	Provide the name and OBI id of the specific method or technology used to generate omics data. Refer to the Controlled Vocabulary file. https://www.ebi.ac.uk/ols4/ontologies/obi/classes/http%253A%252F%252Fpurl.obolibrary.org%252Fobo%252FOBI_0000070?lang=en	whole metagenome sequencing assay, OBI:0002623	M

		<i>Note: This field is mandatory. Enter a null value term from the Controlled Vocabulary file, if information is not available.</i>		
device*	Sample_Information	Name and OBI id of the device(s) used. Refer to the Controlled Vocabulary file. If more than one device was used, list them separated by a semicolon (;). Where possible, the term should be taken from the OBI list of DNA sequencers https://www.ebi.ac.uk/ols4/ontologies/obi/classes/http%253A%252F%252Fpurl.obolibrary.org%252Fobo%252FOBI_04001_03?lang=en <i>Note: This field is mandatory. Enter a null value term from the Controlled Vocabulary file, if information is not available.</i>	PacBio Sequel II,OBI:0002633	M
assembly_software	Sample_Information	Tool(s) used for assembly, including version number and parameters	metaSPAdes 3.11.0;kmer set 21,33,55,77,99,121; default parameters otherwise	X
organism_count	Sample_Information	Total cell count of any organism (or group of organisms) per gram, volume or area of sample, should include name of organism followed by count. The method that was used for the enumeration (e.g. qPCR, atp, mpn, etc.) should also be provided.	total prokaryotes; 3.5e7 cells per ml; qpcr	X
host_symbiont	Sample_Information	The taxonomic name of the organism(s) found living in mutualistic, commensalistic, or parasitic symbiosis with the specific host. The sampled symbiont can have its own symbionts. For example, parasites may have hyperparasites (=parasites of the parasite).	Platyhelminthes, NCBI:txid6157	X
community_type	Sample_Information	Community type of vaginal microbiome	NonLactobacillus, Lcrispatus, Liners, Lgasseri, Ljensenii	X
community_	Sample_Inf	Community state type of vaginal	I	X

state_type	ormation	microbiome - a snapshot of the community at the time of sampling (I,II,III,IV,V)		
subCST	Sample_Inf ormation	Sub-community state type of vaginal microbiome	I-A	X
stability_cat	Sample_Inf ormation	Community stability category based on microbiome composition (Stable/Unstable)	Stable	X
stability_valu e	Sample_Inf ormation	Community stability value based on microbiome composition	0.72591778	X
timepoint_se quential	Sample_Inf ormation	For longitudinal studies, indicate the order in which samples were collected from the same subject and body site. Each sample should be assigned a sequential number (e.g., 1, 2, 3) based on the order of collection. This field helps track the progression of sample collection, and it is recommended to use this field in conjunction with timepoint_numeric, which records the actual time intervals between sample collections.	1	C - Mandatory for longitudinal studies
timepoint_nu meric	Sample_Inf ormation	For longitudinal studies, specify the time elapsed from the start of the experiment till the sample collection. This value should be recorded as an integer, with the corresponding unit specified separately in timepoint_numeric_units (e.g., if the timepoint is "20 weeks", enter timepoint_numeric = 20 and timepoint_numeric_units = weeks). This field provides a precise measure of the time elapsed since the start of the experiment and should be used in conjunction with timepoint_sequential, which tracks the order of sample collection.	20	C - Mandatory for longitudinal studies
timepoint_nu meric_units	Sample_Inf ormation	The unit of measurement associated with timepoint_numeric, in the format of hours, days, weeks, months etc.	weeks	C - Mandatory for longitudinal studies
nugent_scor e	Sample_Inf ormation	Nugent score in the range of 0-10; The Nugent Score is a Gram stain scoring	7	X

		system for vaginal swabs to diagnose bacterial vaginosis.		
nugent_classes	Sample_Information	Based on the Nugent score , provide the Nugent class. 0-3 is considered negative for BV (NO_BV), 4-6 is considered indeterminate for bacterial vaginosis (INTER_BV) , 7-10 is considered indicative of BV (BV); indicative of the absence of lactobacilli and a relative predominance of G. vaginalis and Mobiluncus spp	BV	X
associated_resource	Sample_Information	A related resource that is referenced, cited, or otherwise associated with the sequence.	https://earthmicrobiome.org/	X
sop	Sample_Information	Standard operating procedures used in assembly and/or annotation of genomes, metagenomes or environmental sequences	https://earthmicrobiome.org/protocols-and-standards/its/	X
date_lmp	Enrollment_Information	Date of LMP (Last Menstrual Period) in the format of YYYY-MM-DD	2025-01-01	X
date_usg	Enrollment_Information	Date on which the USG-based dating of pregnancy was done, in the format of YYYY-MM-DD	2025-01-01	X
vdt_enr	Enrollment_Information	Date of enrollment, in the format of YYYY-MM-DD	2025-01-01	X
pregnancy	Enrollment_Information	Date due of pregnancy,calculated at the time of enrollment in the format of YYYY-MM-DD	2025-01-01	X
host	Subject_Information	Name and NCBI taxon ID of the host from which the sample was obtained. Refer to the Controlled Vocabulary file	Homo sapiens,NCBI:txid9606	M
orig_id	Subject_Information	A unique identifier for the subject, assigned by your lab	20-10010	M
host_subject_id	Subject_Information	A unique identifier by which each subject can be referred to, de-identified. Recommended format: 'study_name'_'orig_id'. If unknown, choose 'Not Available' from the pick list.	VMRC_PreSS Mat_20-100010	M
pregnancy_id	Subject_Information	A numerical identifier representing the sequence number of a current pregnancy. This identifier starts at 1 for the first pregnancy and increments by 1	1	X

		for each subsequent pregnancy.		
host_age	Subject_Information	Age of host, in years, at the time of sampling	25	X
host_sex	Subject_Information	Gender or physical sex of the host. <i>Note: This field is mandatory. Enter a null value term from the Controlled Vocabulary file, if information is not available.</i>	F	M
host_body_site	Subject_Information	UBERON term and ID of the body site from where the sample was obtained. Refer to the Controlled Vocabulary file. For Uber-anatomy ontology (UBERON) terms, please see https://www.ebi.ac.uk/ols4/ontologies/uberon/classes/http%253A%252F%252Fpurl.obolibrary.org%252Fobo%252FUBERON_0003134?lang=en	vagina,UBERON:0000996	X
host_body_product	Subject_Information	UBERON term and ID of the substance produced by the body, e.g. Stool, mucus,breast milk, where the sample was obtained from. For Uber-anatomy ontology (UBERON) terms, please see https://www.ebi.ac.uk/ols4/ontologies/uberon/classes/http%253A%252F%252Fpurl.obolibrary.org%252Fobo%252FUBERON_0006314?lang=en	milk,UBERON:0001913	X
host_diet	Subject_Information	Type of diet depending on the host, eg: high-fat, mediterranean etc.; can include multiple diet types	high-fat	X
host_last_meal	Subject_Information	Content of last meal and time since feeding; can include multiple values	french fries; P5H30M	X
host_family_relation	Subject_Information	Relationships to other hosts in the same study; can include multiple relationships	mother;ID298	X
host_genotype	Subject_Information	Observed genotype	ST1	X

host_phenotype	Subject_Information	Phenotype of human or other host. For phenotypic quality ontology (pato) (v 2018-03-27) terms, please see http://purl.bioontology.org/ontology/pato . For Human Phenotype Ontology (HP) (v 2018-06-13) please see http://purl.bioontology.org/ontology/HP	Tinnitus [HP:0000360]	X
sexual_act	Subject_Information	Current sexual partner and frequency of sex		X
douche	Subject_Information	Date of most recent douche in the format of YYYY-MM-DD	2025-01-01	X
birth_control	Subject_Information	Specification of birth control medication used		X
menopause	Subject_Information	Date of onset of menopause in the format of YYYY-MM-DD	2025-01-01	X
hrt	Subject_Information	Whether subject had hormone replacement therapy, and if yes, provide start date in the format of YYYY-MM-DD	yes;2025-01-01	X
hysterectomy	Subject_Information	Specification of whether hysterectomy was performed (yes/no)	no	X
vaginal_pH	Subject_Antropometry	The pH level of vagina	4.4	X
host_tot_mass	Subject_Antropometry	Total mass of the host at sample collection (in kilogram)	65	X
host_height	Subject_Antropometry	The height of subject (in meter)	1.75	X
host_body_mass_index	Subject_Antropometry	Body mass index, calculated as weight/(height)squared (in kilogram per square meter)	22	X
host_pulse	Subject_Antropometry	Resting pulse, measured as beats per minute	65	X
sbp	Subject_Antropometry	Systolic blood pressure of the participant at current antenatal visit	90	X
dbp	Subject_Antropometry	Diastolic blood pressure of the participant at current antenatal visit	60	X

hemo	Subject_Bloodwork	Measurement of hemoglobin concentration in g/dL	measurement value in g/dL	X
hba1c	Subject_Bloodwork	Hemoglobin A1c measurement %	%	X
crp	Subject_Bloodwork	C-reactive protein measurement in mg/dL (1.0-3.0)	mg/dL (1.0-3.0)	X
agp	Subject_Bloodwork	alpha-1-acid glycoprotein measurement in mg/dL (0.6-1.2)	mg/dL (0.6-1.2)	X
ca	Subject_Bloodwork	plasma total calcium concentration measurement in mg/dL (9.0-10.5)	mg/dL (9.0-10.5)	X
fer	Subject_Bloodwork	blood ferritin measurement in ng/mL or ug/L (male:12-300; female:12-150)	ng/mL or ug/L (male:12-300; female:12-150)	X
bun	Subject_Bloodwork	serum urea nitrogen measurement in mg/dL (7-20)	mg/dL (7-20)	X
scr	Subject_Bloodwork	serum creatinine measurement in mg/dL (male:0.6-1.2; female:0.5-1.1)	mg/dL (male:0.6-1.2; female:0.5-1.1)	X
ast	Subject_Bloodwork	serum glutamic-oxaloacetic transaminase measurement in IU/L (10-40)	IU/L (10-40)	X
alb	Subject_Bloodwork	serum albumin measurement in g/dL (3.4-5.4)	g/dL (3.4-5.4)	X
tsh	Subject_Bloodwork	Thyroid-Stimulating Hormone measurement in ug/L (0.4-4.0)	ug/L (0.4-4.0)	X
rpr	Subject_Bloodwork	rapid plasma reagin measurement (positive/negative)	positive/negative	X
sflt	Subject_Bloodwork	soluble fms-like tyrosine kinase-1 measurement in pg/mL or ng/L	pg/mL or ng/L	X
plgf	Subject_Bloodwork	placental growth factor in pg/mL or ng/L	pg/mL or ng/L	X
pappa	Subject_Bloodwork	pregnancy-associated plasma protein-A measurement (MoM or IU/L)	MoM or IU/L	X
vegf	Subject_Bloodwork	vascular endothelial growth factor in pg/mL or ng/L	pg/mL or ng/L	X

HIV_status	Subject_Medical_History	Result of HIV serological test (0 - uninfected/non-reactive; 1- infected/reactive)	0	X
malaria	Subject_Medical_History	Result of malaria test conducted by RDT or smear (0: uninfected/non-reactive; 1: infected/reactive)	0	X
chron_htn	Subject_Medical_History	If the participant has a history of hypertension outside of pregnancy (0 = no; 1 = yes)	0	X
htn_visit	Subject_Medical_History	If the participant had hypertension during the entire pregnancy (0 = no; 1 = yes)	0	X
diabetes	Subject_Medical_History	If the participant has a history of diabetes outside of pregnancy (0 = no; 1 = yes)	0	X
tb	Subject_Medical_History	If the participant has a history of tuberculosis (0 = no; 1 = yes)	0	X
thyroid	Subject_Medical_History	If the participant has a history of hyper- or hypo-thyroidism (0 = no; 1 = yes)	0	X
epilepsy	Subject_Medical_History	If the participant has a history of epilepsy (0 = no; 1 = yes)	0	
host_disease_stat	Subject_Medical_History	List of diseases with which the host has been diagnosed; can include multiple diagnoses. The terms should be chosen from the DO (Human Disease Ontology) at https://www.disease-ontology.org .	measles [DOID:8622]	X
ihmc_medication_code	Subject_Medical_History	Can include multiple medication codes	810	X
chem_administration	Subject_Medical_History	List of chemical compounds administered to the host or site where sampling occurred, and when (e.g. Antibiotics, n fertilizer, air filter); can include multiple compounds. For chemical entities of biological interest ontology (chebi) (v 163), http://purl.bioontology.org/ontology/chebi	agar[CHEBI:2509];2018-05-11 T20:00Z	X
host_body_temp	Subject_Medical_History	Core body temperature of the host when sample was collected (in degree Celsius)	36.5	X

medic_hist_perform	Subject_Medical_History	Whether full medical history was collected (yes/no)	yes	X
perturbation	Subject_Medical_History	Type of perturbation, e.g. chemical administration, physical disturbance, etc., coupled with perturbation regimen including how many times the perturbation was repeated, how long each perturbation lasted, and the start and end time of the entire perturbation period; can include multiple perturbation types	antibiotic addition;R2/2018-05-11T14:30Z/2018-05-11T19:30Z/P1H30M	X
gynecologic_disord	Subject_Medical_History	History of gynecological disorders; can include multiple disorders. The terms should be chosen from the DO (Human Disease Ontology) at http://www.disease-ontology.org , female reproductive system disease (https://disease-ontology.org/?id=DOID:229).		X
urogenit_disord	Subject_Medical_History	History of urogenital disorders, can include multiple disorders. The terms should be chosen from the DO (Human Disease Ontology) at http://www.disease-ontology.org , reproductive system disease (https://disease-ontology.org/?id=DOID:15) or urinary system disease (https://disease-ontology.org/?id=DOID:18).		X
sniff_toba	Subject_Medical_History	If the participant has ever sniffed or chewed tobacco (0=Never, 1=Quit pre-pregnancy, 2=Quit during pregnancy, 3=Currently sniffing/chewing)	0	X
sniff_freq	Subject_Medical_History	If the participant has ever sniffed or chewed tobacco, the frequency per week		
smoke_hist	Subject_Medical_History	Smoking history of the participant (0= Never smoke; 1= Smoked but quit before pregnancy; 2= Quit during pregnancy; 3= Currently smoking)	1	X
passive_smok	Subject_Medical_History	If the participant has lived or currently lives with people who smoke in an enclosed area/room in their presence.(0 = no; 1 = yes)	1	X

alcohol	Subject_Medical_History	If the participant drinks alcohol (0=Never, 1=Yes, quit pre-pregnancy, 2=Yes, quit during pregnancy, 3=Currently drinking)	1	X
alcohol_freq	Subject_Medical_History	If the participant drinks alcohol, the average number of units consumed per week (1 unit = small glass (125 ml) of wine or one bottle/can (330 ml) of beer)		X
vag_bleed	Subject_Medical_History	If any vaginal bleeding reported by the participant since last visit (0 = no; 1 = yes)	0	X
swell_ankle	Subject_Medical_History	If any swelling in the ankle of the participant (0 = no; 1 = yes)	0	X
syphilis	Subject_Medical_History	Result of Syphilis testing at enrollment (0 = non-reactive/negative; 1 = reactive/positive; 2 = indeterminate)	0	X
udip_prot	Subject_Medical_History	Semi quantitative urine test strip analysis (0 = neg; 1 = trace; 2 = positive)	1	X
gravidity	Subject Obstetric_History	Total number of times the participant has been pregnant, including the current pregnancy	3	X
parity	Subject Obstetric_History	Number of pregnancies the participant has carried to at least 24 weeks	2	X
prev_mis	Subject Obstetric_History	Number of previous pregnancies the participant has had that ended before 6 months	1	X
prev_sb	Subject Obstetric_History	If the participant has ever had stillbirth at a previous delivery (0 = no; 1 = yes)	1	X
prev_sb_no	Subject Obstetric_History	Number of previous deliveries the participant has had of a stillborn child	1	X
prev_lb	Subject Obstetric_History	Number of deliveries the participant has had of a live child	2	X
prev_ptb	Subject Obstetric_History	If the participant has ever had a baby that was born preterm (<37 weeks GA) (0 = no; 1 = yes)	1	X

prev_ptb_no	Subject_Obstetric_History	Number of previous pregnancies the participant has had that ended before term (<37 weeks GA)	1	X
prev_ptb_multi	Subject_Obstetric_History	Number of previous preterm births (<37 weeks GA) in which the participant had more than 1 baby	0	X
prev_multip	Subject_Obstetric_History	If the participant delivered twins or triplets in any of her previous pregnancies(0 = no; 1 = yes)	0	X
prev_cs	Subject_Obstetric_History	If the participant had a caesarean section in any of the previous pregnancies (0 = no; 1 = yes)	0	X
alive_child	Subject_Obstetric_History	Number of living children the participant has	2	X
birth_interval	Subject_Obstetric_History	Interval between the participant's previous pregnancy outcome and the last menstrual period for the current pregnancy (in months)	18	X
race	Socioeconomic_Determinants	A category of people who identify with each other, usually on the basis of presumed similarities such as a common language, ancestry, history, society, culture, nation or social treatment within their residing area. https://en.wikipedia.org/wiki/List_of_contemporary_ethnic_groups <i>Note: This field is mandatory. Enter a null value term from the Controlled Vocabulary file, if information is not available.</i>	Asian,Black,White,Latino,Not Available	M
host_occupation	Socioeconomic_Determinants	Occupation of the participant (0 = not employed; 1 = employed)	0	X
host_education	Socioeconomic_Determinants	Total number of years of completed schooling/education of the participant	10	X
husband_education	Socioeconomic_Determinants	schooling /education of the participant's husband	10	X

Table 2. MIMAG-specific metadata attributes

Attribute	Group	Description	Example	Mandatory(M) /Optional(O) /Conditional Mandatory (C)
tax_ident	Sample_Information	The phylogenetic marker(s) used to assign an organism name to the SAG or MAG	rpoB gene	X
assembly_qual	Sample_Information	The assembly quality category is based on sets of criteria outlined for each assembly quality category. For MIMAG; Finished: Single, validated, contiguous sequence per replicon without gaps or ambiguities with a consensus error rate equivalent to Q50 or better. High Quality Draft: Multiple fragments where gaps span repetitive regions. Presence of the 23S, 16S and 5S rRNA genes and at least 18 tRNAs. Medium Quality Draft: Many fragments with little to no review of assembly other than reporting of standard assembly statistics. Low Quality Draft: Many fragments with little to no review of assembly other than reporting of standard assembly statistics. Assembly statistics include, but are not limited to total assembly size, number of contigs, contig N50/L50, and maximum contig length.	High-quality draft genome	X
assembly_name	Sample_Information	Name/version of the assembly provided by the submitter that is used in the genome browsers and in the community	HuRef, JCVI_ISG_i3_1.0	X
annot	Sample_Information	Tool used for annotation, or for cases where annotation was provided by a community jamboree or model organism database rather than by a specific submitter	prokka	X
number_contig	Sample_Information	Total number of contigs in the cleaned/submitted assembly that makes up a given genome, SAG, MAG, or UViG	40	X
feat_pred	Sample_Information	Method used to predict UViGs features such as ORFs, integration site, etc.	Prodigal 2.6.3	X
ref_db	Sample_Information	List of database(s) used for ORF annotation, along with version number and reference to website or publication	pVOGs 5; http://dmk-brain.ecn.uiowa.edu/pVOGs/ Grazziotin et al. 2017 doi:10.1093/nar	X

			/gkw975	
sim_search_meth	Sample_Information	Tool used to compare ORFs with database, along with version and cutoffs used	HMMER3 3.1b2; hmmsearch, cutoff of 50 on score	X
tax_class	Sample_Information	Method used for taxonomic classification, along with reference database used, classification rank, and thresholds used to classify new genomes	vContact2 0.9.19 (references from NCBI RefSeq v83, genus rank classification, default parameters)	X
x16s_recover	Sample_Information	Can a 16S gene be recovered from the submitted SAG or MAG?	yes	X
x16s_recover_software	Sample_Information	Tools used for 16S rRNA gene extraction	rambl v2; default parameters	X
trnas	Sample_Information	The total number of tRNAs identified from the SAG or MAG	18	X
trna_ext_software	Sample_Information	Tools used for tRNA identification	infernol v2; default parameters	X
compl_score	Sample_Information	Completeness score is typically based on either the fraction of markers found as compared to a database or the percent of a genome found as compared to a closely related reference genome. High Quality Draft: >90%, Medium Quality Draft: >50%, and Low Quality Draft: < 50% should have the indicated completeness scores	Medium Quality Draft : 60%	X
compl_software	Sample_Information	Tools used for completion estimate, i.e. checkm, anvi'o, busco	checkm	X
compl_appr	Sample_Information	The approach used to determine the completeness of a given genomic assembly, which would typically make use of a set of conserved marker genes or a closely related reference genome.	UViG length compared to the average length of reference genomes from the P22virus genus (NCBI RefSeq v83)	X
contam_sco	Sample_Inf	The contamination score is based on the	0.01	X

re	ormation	fraction of single-copy genes that are observed more than once in a query genome. The following scores are acceptable for; High Quality Draft: < 5%, Medium Quality Draft: < 10%, Low Quality Draft: < 10%. Contamination must be below 5% for a SAG or MAG to be deposited into any of the public databases		
contam_scr een_input	Sample_Inf ormation	The type of sequence data used as input	contigs	X
contam_scr een_param	Sample_Inf ormation	Specific parameters used in the decontamination software, such as reference database, coverage, and kmers. Combinations of these parameters may also be used, i.e. kmer and coverage, or reference database and kmer	kmer	X
decontam_s oftware	Sample_Inf ormation	Tool(s) used in contamination screening	anvi'o	X
bin_param	Sample_Inf ormation	The parameters that have been applied during the extraction of genomes from metagenomic datasets	coverage and kmer	X
bin_softwar e	Sample_Inf ormation	Tool(s) used for the extraction of genomes from metagenomic datasets, where possible include a product ID (PID) of the tool(s) used.	MetaCluster-TA (RRID:SCR_00 4599)	X
reassembly _bin	Sample_Inf ormation	Has an assembly been performed on a genome bin extracted from a metagenomic assembly?	no	X
mag_cov_s oftware	Sample_Inf ormation	Tool(s) used to determine the genome coverage if coverage is used as a binning parameter in the extraction of genomes from metagenomic datasets	bbmap	X

Table 3. MIMARKS-specific metadata attributes

Attribute	Group	Description	Example	Mandatory(M) /Optional(O) /Conditional Mandatory (C)
target_gene	Sample_Inf	Targeted gene or locus name for marker	16S rRNA,	M

	ormation	gene studies <i>Note: This field is mandatory. Enter a null value term from the Controlled Vocabulary file, if information is not available.</i>	18S rRNA, nif, amoA, rpo	
target_subfragment	Sample_Information	Name of subfragment of a gene or locus. Important to e.g. identify special regions on marker genes like V6 on 16S rRNA. <i>Note: This field is mandatory. Enter a null value term from the Controlled Vocabulary file, if information is not available.</i>	V6, V9, ITS	M
pcr_primers	Sample_Information	PCR primers that were used to amplify the sequence of the targeted gene, locus or subfragment. This field should contain all the primers used for a single PCR reaction if multiple forward or reverse primers are present in a single PCR reaction. The primer sequence should be reported in uppercase letters. <i>Note: This field is mandatory. Enter a null value term from the Controlled Vocabulary file, if information is not available.</i>	FWD:GTGCC AGCMGCCGC GGTAA; REV:GGACTA CHVGGGTWT CTAAT	M
pcr_cond	Sample_Information	Description of reaction conditions and components of PCR in the form of 'initial denaturation:94degC_1.5min; annealing=...'	initial denaturation:94_3; annealing:50_1; elongation:72_1.5;final elongation:72_10	X
seq_quality_check	Sample_Information	Indicate if the sequence has been called by automatic systems (none) or undergone a manual editing procedure (e.g. by inspecting the raw data or chromatograms). Applied only for sequences that are not submitted to SRA,ENA or DRA	none	X
chimera_check	Sample_Information	Tool(s) used for chimera checking, including version number and parameters, to discover and remove chimeric sequences. A chimeric sequence consists of two or more phylogenetically distinct parent sequences.	uchime v4.1	X

B. Metadata Validation using DataHarmonizer

[DataHarmonizer](#) is a standardized browser-based spreadsheet editor and validator application, created by the Centre for Infectious Disease Genomics and One Health (CIDGOH) at Simon Fraser University. We have developed two custom-made templates for the VMRC project, to be used with DataHarmonizer, for data validation purposes.

Steps to use DataHarmonizer

1. Download web.zip package from the Google drive ([link](#))
2. Unzip the package.
3. Get the path to where the web package got downloaded in your system and open the web/dist/index.html in your web browser like:
file:///Users/jane/Downloads/web/dist/index.html

This should open up the DataHarmonizer validator spreadsheet, like below (Fig. 1).

File

Settings

Validate

Help

Template

CanCOGeN Covid-19

Loaded file

Database Identifiers		Sample collection and processing					
	specimen collector sample ID	GenBank accession	GISAID accession	sample collected by	sample collector contact email	sample collector contact address	sequence submitted by
1				▼			▼
2				▼			▼
3				▼			▼
4				▼			▼
5				▼			▼
6				▼			▼
7				▼			▼
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96				▼			▼
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100				▼			▼

Add

100

more rows at the bottom.

Fig. 1: DataHarmonizer Template Spreadsheet

Filling in the spreadsheet and validating your metadata

1. Select the template you want to use for your metagenome or 16S dataset by selecting the “vmrc_MIMAG/vmrc_mimag” or “vmrc_MIMARKS/vmrc_mimarks” from the **Template** menu beside the **Help** button.

Data can be entered into the validator application manually, by typing values into the application’s spreadsheet, or data can be imported from local xlsx, xls, tsv and csv files.

To import local data, click **File** on the top-left toolbar, and then click **Open**. To enter data in a new file, click **File** on the top-left toolbar, and then click **New**. Data entered into the spreadsheet can be copied and pasted.

Note: Only files containing the headers expected by the DataHarmonizer can be opened in the application. So please make sure to follow the metadata structure as in the example ones.

2. Familiarize yourself with DataHarmonizer functionality by reviewing the “**Getting Started**”. To access "Getting Started", click on the green **Help** button on the top-left toolbar, then click **Getting Started**. Definitions, examples and further guidance are available by double clicking on the field headers, or by using the “**Reference Guide**”. To access the “Reference Guide” click on the **Help** button, then click **Reference Guide**. You can load the example input files in the exampleInput folders within each template directory of the DataHarmonizer web package (e.g., web/templates/vmrc_MIMAG/exampleInput and web/templates/vmrc_MIMARKS/exampleInput).
3. Enter data into the validator spreadsheet.

Fields are color-coded; required fields (minimal metadata) are yellow, strongly recommended fields are purple, optional fields are white/gray.

- Hide non-required fields (color-coded purple and white/grey) by clicking **Settings** on the top-left toolbar, followed by clicking on **Show Required Columns** (color-coded in yellow).
 - Double click on the field headers to see definitions and detailed guidance as needed
 - Jump to a specific field header by clicking **Settings** on the top-left toolbar, followed by clicking on **Jump to**, then select the field header of the column you would like to view from the drop down list.
 - Populate the validator template with the information from your dataset.
 - Use picklists of controlled vocabularies when provided.
 - A value must be entered for every required field (yellow colored ones) in each row. If data is missing or not collected, **choose a null value from the picklist or the Controlled Vocabularies file** .
 - **Not Applicable** - For conditional mandatory fields, where conditions are not met
 - **Not Available** - Required field, but information is missing, not collected or not provided
 - Free text can be provided when picklists are not available.
- Optional fields can be left blank
- For filling an entire column with the same data, use the **Fill Column** function. Click **Settings**, followed by **Fill Column**. Type in the name of the desired field, followed by the value that should be used to fill every row in that column. Then click **OK**.

If a desired term is not present in the picklist, please contact Jain Aluvathingal at jgeorge@som.umaryland.edu

***Note:** Sometimes there will be constraints on what information can be shared, other times a field may not be applicable to your sample. Use the null values in the picklist to report missing data.*

4. Validate the entered data by clicking on the **Validate** button on the top-left toolbar.

Missing information and invalid entries in required fields will be highlighted in red.

- Observe invalid rows by clicking **Settings** in the top-left toolbar, and then clicking on **Show invalid rows**.
- Navigate to and between errors by clicking the **Next Error** button that appears when errors are present. After resolving these errors, hit the **Validate** button again to see if any errors remain. If there are no more errors the **Next Error** button will change to **No Errors** and fade away.
- Observe valid rows by clicking **Settings** in the top-left toolbar, and then clicking on **Show valid rows**.
- Return view to all rows by clicking **Settings** in the top-left toolbar, and then clicking on **Show all rows**.

Note: Row viewing options only appear after a validation attempt has been made.

5. Address any invalid data that was flagged in red in the template.
 - Pale Red = Incorrect data format (spelling errors, MM-DD-YYYY format error)
 - Dark Red = Required data missing
6. Once all errors are corrected and the metadata is successfully validated, the metadata file is considered harmonized and ready for upload to public or private repositories. You can save the validated metadata in various formats—such as *.xlsx*, *.xls*, *.tsv*, *.csv*, or *.json*—by clicking **File** and then selecting **Save As**. Enter the desired file name and click **Save**.

If you wish to export the validated metadata in BioSample format, click **File** in the top-left toolbar, then select **Export To** and choose the appropriate option.

Note: It is possible to export incomplete or invalid data. Make sure to review any errors prior to exporting.

7. If your template uses the DataHarmonizer provenance field, provenance information will automatically be added upon “Validation” for all rows containing data. Provenance information includes the DataHarmonizer application version number, and the template schema version number (if available).

C. Generic directory structure for VMRC data

All data files are organized under a “parent” folder, typically representing a project or a study. Please refer to the generic directory structure shown below (Fig. 2) for guidance on how to store your files and metadata in the data storage platform.

You need to create folders **only** for the datatypes you have. For instance, if you have only metagenome and metatranscriptome data, there is no need to create folders for 16S amplicon, immunology, microbial genome, metabolome, or host genome data.

<Study_name> as top-level folder

- 16S_metadata_file**
- MIMAG_metadata_file**
- 16Samplicon**
 - FASTA
 - job_submission_script
 - summary_files (Abundance tables)
 - counts_by_ASV
 - counts_by_TAXON
 - abundance_table_StR_CST.csv
 - sequence_reads
 - v3v4 (names of files should reflect v3v4, etc)
 - v4
 - PacBio

--ITSamplicon

- FASTA
- sequence_reads
 - v3v4 (names of files should reflect v3v4, etc)
 - v4
 - PacBio
- summary_files (Abundance tables)
 - counts_by_ASV
 - counts_by_TAXON
 - abundance_table_StR_CST.csv

--Immunology

- summary_files

--MicrobialGenome

- assemblies
 - PacBio
 - Illumina
 - Nanopore
- sequence_reads
 - PacBio
 - Illumina
 - Nanopore
- annotation
 - prokka
- summary_files

--Metabolome

- summary_files

--HostGenome

- assemblies
 - PacBio
 - Illumina
 - Nanopore
- sequence_reads
 - PacBio
 - Illumina
 - Nanopore
- summary_files

--Metagenome

- assemblies
 - PacBio
 - Illumina
 - Nanopore
- sequence_reads

```
--PacBio
--Illumina
--Nanopore
--summary_files

--Metatranscriptome
--human_transcriptome
--assemblies
--PacBio
--Illumina
--Nanopore
--sequence_reads
--PacBio
--Illumina
--Nanopore
--summary_files
--microbial_transcriptome
--assemblies
--PacBio
--Illumina
--Nanopore
--sequence_reads
--PacBio
--Illumina
--Nanopore
--summary_files
```

Fig. 2: Generic directory structure for VMRC data